

Knowledge Article Retrieval for a ServiceNow AI Agent

Concepts, configuration, and running the agent

1. The concepts

RAG (Retrieval Augmented Generation) grounds an LLM answer in your own data. In ServiceNow it is native, built on AI Search. Three steps: **Retrieval** finds the relevant Knowledge articles the user is allowed to see, **Augmentation** passes that article content plus the question to the LLM, and **Generation** produces an answer grounded in the article, with a link back to the source. Content security and ACLs are enforced during retrieval automatically, so an agent never grounds on articles the user cannot access.

Under the hood, AI Search is a layered hierarchy. This is the chain your configuration sets up:

Search Application → **Search Profile** → **Search Source** → **Indexed Source** → **Semantic Index Config** → **Semantic Field**

Layer	What it does for Knowledge retrieval
Search Source	Points at the kb_knowledge table, the data the agent searches.
Indexed Source	Marks that source as enabled for semantic or hybrid search.
Semantic Index Config	Controls vector embeddings and chunking, the heart of semantic search.
Semantic Field	The article fields that get embedded, for example Short description and the article body.
Search Profile	Defines which sources are in scope and how results rank. This is what the agent calls.

Three search modes worth knowing before you configure:

- **Keyword (lexical)**: literal term matches, fast and precise, but misses paraphrased questions.
- **Semantic vector**: matches on meaning using embeddings, so different wording still finds the right article. Auto-enabled, no knobs of its own.
- **Hybrid**: runs both and blends the scores. This is what you want for accurate Knowledge retrieval.

2. Configure Knowledge article retrieval

1 Set up the Search Source on Knowledge

- Go to AI Search and open Search Sources. Confirm a source on the **kb_knowledge** table exists, or create one.
- Make sure the Knowledge Base you want to expose is in scope, and that the articles are **published**. Drafts and retired articles will not retrieve.

2 Enable the Indexed Source and Semantic Index

- Open the Indexed Source for Knowledge and confirm it is enabled for **semantic or hybrid** search.
- In the Semantic Index Config, set the **Semantic Fields** to the fields that carry meaning, typically **Short description** and the **article body (text)**.
- Set chunking: **chunking unit** (words or sentences), **chunk size** (default around 250 words or 15 sentences), and choose **fixed-size** for most KBs or **small-to-big** for long articles.
- Build or refresh the index. Embeddings must exist before anything retrieves.

3 Create the Search Profile

- Create a Search Profile scoped to the Knowledge source only. Tight scope means less noise and higher accuracy.
- Confirm it uses **hybrid** behaviour so keyword and semantic signals both contribute.

4 Validate in Preview Search first

- Open AI Search, Preview Search console, select your profile, and run a natural language question phrased differently from the article wording.
- Confirm the right article comes back. If retrieval is wrong here, fix it now. It will be wrong in the agent otherwise.

Prerequisite: semantic and hybrid AI Search that feeds Now Assist requires Now Assist / AI Search entitlement and is not available on a free PDI. Field labels also shift slightly between releases, so verify against your instance.

3. Add retrieval to the AI Agent

5 Open the agent and add a Search Retrieval tool

- In AI Agent Studio, open or create your agent and go to **Add information**, then **Add a search retrieval**.
- **Name**: a clear tool name, for example Search knowledge articles.
- **Description (AI instruction)**: tell the agent when to use it, for example: Use this to retrieve Knowledge articles when the user asks how to do something or for troubleshooting steps. The agent reads this to decide when to call the tool, so be specific.
- **Search profile**: select the Knowledge profile from Step 3.
- **Search sources**: confirm **Knowledge** is the source in scope.
- **Fields returned**: return only what the answer needs, for example Short description, article body, and the KB number for citation.
- **Results limit**: keep it tight, 3 to 5. Too many results dilutes the prompt and invites hallucination.
- Save the tool and the agent.

4. Run the agent

6 Test and confirm grounding

- Open the agent test panel and ask a natural language Knowledge question, ideally worded differently from the article.
- Confirm the answer is grounded in the actual article and cites the KB source. That grounding plus citation is the whole point of RAG.
- Ask something the Knowledge Base does not cover and confirm the agent declines rather than inventing an answer.
- If results are off, return to Preview Search to debug retrieval, then tune Semantic Fields, chunking, profile scope, fields returned, or results limit.

Quick recall: index the right fields, chunk sensibly, scope the profile to Knowledge, use hybrid, return few fields, keep the results limit low, and write a precise tool description. Those are the levers that move accuracy.